1	(a)	Define *gravitational potential at a point*.		
		.....		
		.....		[1]
	L1	It is the **work done per unit mass by an external force in bringing a small test mass from infinity to that point** (without a change in kinetic energy).		B1
	(b)	A satellite of mass m orbits a planet of mass M and radius R_p. The radius of the orbit is R. The satellite and the planet may be considered to be point masses with their masses concentrated at their centres. They may be assumed to be isolated in space. For the satellite in its orbit, show that		
		(i)	its kinetic energy E_k is given by $E_k = \frac{GMm}{2R}$,	
				[2]
	L1	The **centripetal force (F_C) on the satellite is provided by the gravitational force (F_g)**. $F_g = F_c$ $\frac{GMm}{R^2} = \frac{mv^2}{R}$ $mv^2 = \frac{GMm}{R}$ $\frac{1}{2}mv^2 = \frac{1}{2}\frac{GMm}{R}$ $E_k - \frac{1}{2}mv^2$ $= \frac{GMm}{2R}$ (*shown*)	B1 M1 A0	
		(ii)	the ratio $\frac{\text{kinetic energy of satellite } E_k}{\text{total energy of satellite } E_T}$ is equal to -1.	

				[2]
		L1	$E_P = -\frac{GMm}{R}$ $E_T = E_P + E_k$ $= -\frac{GMm}{R} + \frac{GMm}{2R}$ $= -\frac{GMm}{2R}$ $\therefore \frac{E_k}{E_T} = \frac{\frac{GMm}{2R}}{-\frac{GMm}{2R}}$ $= -1 (shown)$ ***Candidates must show how they obtain E_T.***	M1 M1 A0
	(c)	The satellite in **(b)** gradually loses energy due to small resistive forces. Suggest why many such satellites eventually 'burn up' in the Earth's atmosphere.		
		.....		
		.....		
		.....		
		.....		
		.....		
		.....		
		.....		[3]
		L3	As the satellite gradually loses energy, i.e. its total energy E_T decreases (i.e. more negative), its radius of orbit R decreases. As the satellite radius of orbit R decreases, its kinetic energy E_k increases, i.e. its	B1 B1

		speed *v* increases as it moves nearer or into the Earth's atmosphere. The satellite's increasing speed gives rise to increase in resistive force acting on the satellite, which results in increasing rate of conversion of its energy to thermal energy, and so the satellite could eventually 'burn up' in the Earth's atmosphere.	**B1**

[Total: 8]